Lab Code	Lab Name	Credit
CSDL7013	**Natural Language processing Lab**	**1**

Prerequisite: Java/Python	
Lab Objectives: The course aims	
1	To understand the key concepts of NLP.
2	To learn various phases of NLP.
3	To design and implement various language models and POS tagging techniques.
4	To understand various NLP Algorithms
5	To learn NLP applications such as Information Extraction, Sentiment Analysis, Question answering, Machine translation etc.
6	To design and implement applications based on natural language processing
Lab Outcomes: Learners will be able	
1	Apply various text processing techniques.
2	Design language model for word level analysis.
3	Model linguistic phenomena with formal grammar.
4	Design, implement and analyze NLP algorithms.
5	To apply NLP techniques to design real world NLP applications such as machine translation, sentiment analysis, text summarization, information extraction, Question Answering system etc.
6	Implement proper experimental methodology for training and evaluating empirical NLP systems.

Suggested List of Experiments **(Select a case study and perform the experiments 1 to 8.)**.	
Star (*) marked experiments are compulsory.	
Sr. No.	**Name of the Experiment**
1	Study various applications of NLP and Formulate the Problem Statement for Mini Project based on chosen real world NLP applications: [Machine Translation, Text Categorization, Text summarization, chat Bot, Plagarism, Spelling & Grammar checkers, Sentiment / opinion analysis, Question answering, Personal Assistant, Tutoring Systems, etc.]
2	Apply various text preprocessing techniques for any given text : Tokenization and Filtration & Script Validation.
3	Apply various other text preprocessing techniques for any given text : Stop Word Removal, Lemmatization / Stemming.
4	Perform morphological analysis and word generation for any given text.
5	Implement N-Gram model for the given text input.
6	Study the different POS taggers and Perform POS tagging on the given text.
7	Perform Chunking for the given text input.
8	Implement Named Entity Recognizer for the given text input.

9	Implement Text Similarity Recognizer for the chosen text documents.
10	Exploratory data analysis of a given text (Word Cloud)
11	Mini Project Report: For any one chosen real world NLP application.
13	Implementation and Presentation of Mini Project

Term Work:	
1	Study various applications of NLP and Formulate the Problem Statement for Mini Project based on chosen real world NLP applications: [Machine Translation, Text Categorization, Text summarization, chat Bot, Plagarism, Spelling & Grammar checkers, Sentiment / opinion analysis, Question answering, Personal Assistant, Tutoring Systems, etc.]
2	Apply various text preprocessing techniques for any given text: Tokenization and Filtration & Script Validation.
3	Apply various other text preprocessing techniques for any given text: Stop Word Removal, Lemmatization / Stemming.